

05 — Limitations

Problem-statement deliverable: *Documentation explaining limitations of the system*

Estimates and visuals are planning aids. Treat them as approximate.

1. Advisory costing (explicit)

- Outputs are **not** legally binding quotations or contractor bills.
- Rates are catalogue defaults or user edits — not live market surveys.
- Labour is folded into a single blended `rate_inr` per material, not a separate labour schedule.
- Waste is a flat **10%** factor, not site-specific wastage.

2. Area & quantity accuracy

Limitation	Impact
User must enter facade W×H	Wrong dimensions scale every line item
Mask fraction = area fraction	Perspective / depth / foreshortening not corrected
No door-height or window-scale reference	Automatic metric scale unavailable
Single photo	Only visible surfaces; sides/back omitted
Unit categories use instance counts	Missed or merged detections skew piece counts
No paint L / tile piece / railing RFT packing	Quantities are sqm or pcs only

Improve accuracy by correcting masks carefully and measuring the real facade width/height with a tape.

3. Segmentation

- Grounding DINO can miss unusual architecture, heavy occlusion, night photos, or cluttered scenes.
- Wall quality depends on building box quality and opening subtraction.
- Soft edges / vegetation may bleed into masks — use brush/eraser.
- Parapet may appear as part of wall or rooftop depending on prompt hits.
- CPU-only; **cold start is slow**; unsuitable for high-concurrency production without a job queue.

4. Visualization / rendering

- Rendering is **classical OpenCV** (recolor + texture tiling), not photorealistic diffusion.
- Textures tile approximately; fine mortar joints and specular materials will not match showroom photos.
- Lighting approximation preserves luminance shading but is not full global illumination.
- ControlNet / Stable Diffusion paths exist conceptually in earlier design docs but are **disabled** (memory / RAM budget ~8 GB).
- Structure preservation is strong for paint; texture placement can still look “stamped” on complex geometry.

5. Product / platform gaps

Gap	Note
No authentication	Local single-user demo
No multi-design named variants	Last render overwrites output image
No multi-photo / 360 capture	One elevation at a time
Building-presence rejection soft	Severe non-house images may still upload; quality warnings only
Region UX	Brush/erase yes; full redraw/rename/delete region polish is limited

6. Assumptions that can fail

- Shot is approximately frontal
- Building is low-rise residential exterior
- Masks after edit represent surfaces that will actually be finished
- User rates reflect local market

If these fail, both visualization quality and cost credibility degrade.

7. Honest positioning

Strength	Bound
Fast visual “what if” on the user’s own photo	Not a construction drawing
Transparent qty × rate math	Not a surveyor’s BOQ

Editable rates + PDF for contractor talks	Not a signed quotation
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The problem statement asked for **reasonably accurate** area for costing purposes. This prototype meets that bar when the user calibrates facade size and corrects masks; it does **not** claim centimetre survey accuracy.